Experiment No: 5

ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;
void setup() {
//configure pin2 as an input and enable the internal pull-up resistor
pinMode(12, INPUT_PULLUP);
pinMode(1,OUTPUT);
  pinMode(2,OUTPUT);
  pinMode(3,OUTPUT);
  pinMode(4,OUTPUT);
  pinMode(5,OUTPUT);
  pinMode(6,OUTPUT);
  pinMode(7,OUTPUT);

}
void loop() {
  //read the pushbutton value into a variable
  int sensorVal = digitalRead(12);

    if (sensorVal == LOW) {
       x0=true;
    }
      while(x0){
        zero();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x1=true;
              x0=false;
            }
               }
      while(x1){
        one();
          sensorVal = digitalRead(12);
```

```
          if (sensorVal == LOW) {
              x2=true;
              x1=false;
            }
               }
       while(x2){
        two();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x3=true;
              x2=false;
            }
               }
     while(x3){
        three();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x4=true;
              x3=false;
            }
               }
       while(x4){
        four();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x5=true;
              x4=false;
            }
               }
       while(x5){
        five();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x6=true;
              x5=false;
            }
               }
     while(x6){
        six();
          sensorVal = digitalRead(12);
          if (sensorVal == LOW) {
              x7=true;
              x6=false;
```

```
             }
                }
        while(x7){
         seven();
           sensorVal = digitalRead(12);
           if (sensorVal == LOW) {
               x8=true;
               x7=false;
             }
                }
     while(x8){
         eight();
           sensorVal = digitalRead(12);
           if (sensorVal == LOW) {
               x9=true;
               x8=false;
             }
                }
    while(x9){
         nine();
           sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}
}
void zero()
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
}
void one()
{
}
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
```

Experiment No: 5

ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

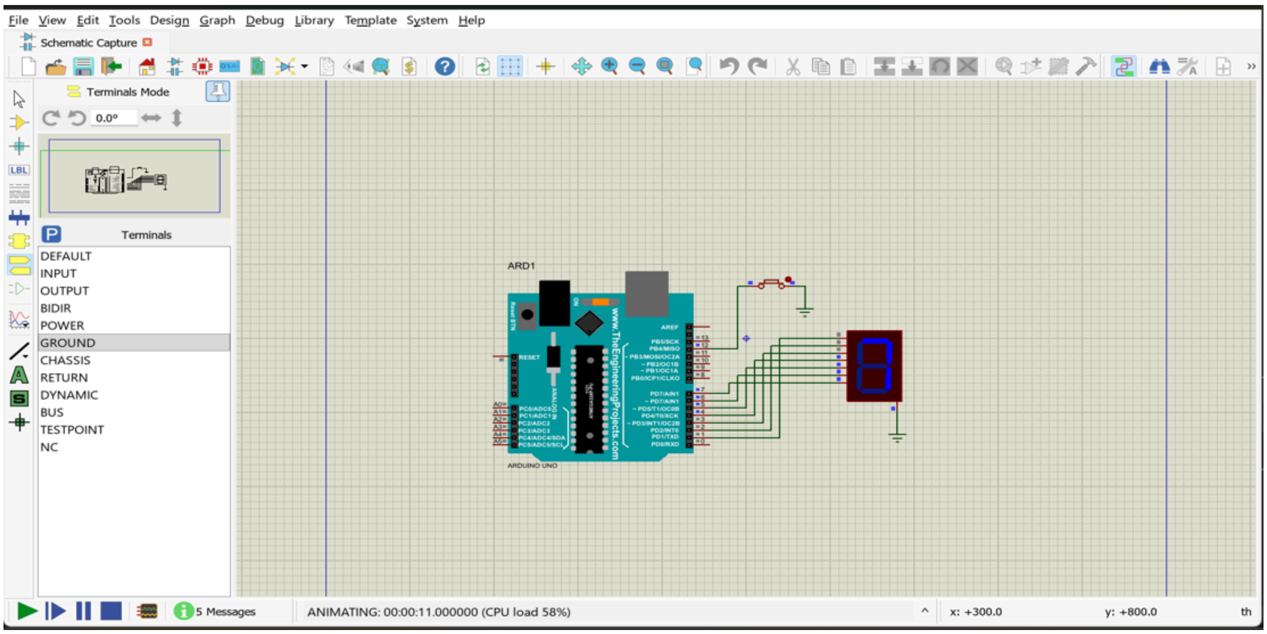
```
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
void two()
{
}
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
void three()
{
}
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
void four()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void five()
{
}
{
}
digitalWrite(1,HIGH);
```

```
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
void six()
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
void seven()
{
}
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
void eight()
{
}
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
```

Experiment No: 5
ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

```
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

OUTPUT :



RESULT: Thus the program has been successfully verified and executed.